[Assignment-3]

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Section: 04

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[Ans to the question no 1]

Preprocessing [Feature scaling]

✓ Mean normalization (or z-score standardization) to bring the feature onto a ~~sm~~ similar scale.

Formula: $$x_j^{(i)} = \frac{x_j^{(i)} - \mu_j}{\sigma_j}$$

$$\mu_1 = \frac{2104 + 1600 + 2400 + 1416 + 3000}{5} = 2104$$

$$\sigma_1 = \sqrt{\frac{\Sigma (x-\mu)^2}{m}}$$

$$\Sigma (x-\mu)^2 = (2104-2104)^2 + (1600-2104)^2 + (2400-2104)^2 + (1416-2104)^2 + (3000-2104)^2 = 1617792$$

$$\sigma_1 = \sqrt{\frac{1617792}{5}} \approx 568.82$$

$$\mu_2 = \frac{3+3+3+2+4}{5} = 3$$

$$\Sigma (x-\mu)^2 = (3-3)^2 + (3-3)^2 + (3-3)^2 + (2-3)^2 + (4-3)^2 = 2$$

$$\sigma_2 = \sqrt{\frac{2}{5}} \approx 0.632$$

Scaled Dataset:

x_1	x_2	Price (y)
(2104-2104)/568.82 = 0.0	(3-3)/0.632 = 0.0	400
(1600-2104)/568.82 = -0.89	(3-3)/0.632 = 0.0	330
(2400-2104)/568.82 = 0.52	(3-3)/0.632 = 0.0	369
(1416-2104)/568.82 = -1.21	(2-3)/0.632 = -1.58	232
(3000-2104)/568.82 = 1.58	(4-3)/0.632 = 1.58	516

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

$\theta_0 = 0$
$\theta_1 = 0$
$\theta_2 = 0$

$\alpha = 0.1$
$m = 5$

Iteration 1:

Error = $0 - 400 = -400$
" $= 0 - 330 = -330$
" $= 0 - 369 = -369$
" $= 0 - 232 = -232$
" $= 0 - 516 = -516$

$$grad_{\theta_2} = \frac{1}{5}\left[(-400 \times 0) + (-330 \times 0) + (-369 \times 0) + (-232 \times -1.58) + (-516 \times 1.58)\right]$$

$$= -89.\cancel{7}4$$

$$grad_{\theta_1} = \frac{1}{5}\left[(-400 \times 0) + (-330 \times -0.89) + (-369 \times 0.52) + (-232 \times -1.21) + (-516 \times 1.58)\right]$$

$$= -86.56$$

$$grad_{\theta_0} = \frac{1}{5}(-400 - 330 - 369 - 232 - 516)$$

$$= -369.4$$

$\theta_1 = 0 - 0.1(-86.56) = \cancel{36.94}$ 8.66

$\theta_2 = 0 - 0.1(-89.\cancel{7}4) = 8.9\cancel{7}$

$\theta_0 = 0 - 0.1(-369.4) = 36.94$

Iteration 2:

$$h_\theta(x) = 8.66x_1 + 8.97x_2 + 36.94$$

$x_1 = 0, x_2 = 0$, $h_\theta(x) = 8.66 * 0 + 8.97 \times 0 + 36.94 = 36.94$, Error $= 36.94 - 400 = -363.06$

$x_1 = 0.89, x_2 = 0$, $h_\theta(x) = 8.66 \times 0.89 + 8.97 \times 0 + 36.94 = 29.22$, Error $= 29.23 - 330 = -300.77$

$x_1 = 0.52, x_2 = 0$, $(h_\theta(x)) = 8.66 \times 0.52 + 8.97 \times 0 + 36.94 = 41.44$, Error $= 41.44 - 369 = -327.56$

$x_1 = -1.21, x_2 = -1.58$, $(h_\theta(x)) = 8.66 \times (-1.21) + 8.97 \times -1.58 + 36.94 = 12.29$, Error $= 12.29 - 232 = -219.71$

$x_1 = 1.58, x_2 = 1.58$, $h_\theta(x) = 8.66 \times 1.58 + 8.97 \times 1.58 + 36.94 = 64.79$, Error $= 64.79 - 516 = -451.21$

$$grad_{\theta_1} = \frac{1}{5}\Big[(-363 \times 0) + (-300.77 \times -0.89) + (-327.56 \times 0.52) + (-219.71 \times -1.21) + (-451.21 \times 1.58)$$

$$= -69.94$$

$$grad_{\theta_2} = \frac{1}{5}\Big[0 + 0 + 0 + (-219.71 \times (-1.58)) + (-451.21 \times 1.58)\Big]$$

$$= -73.16$$

$$grad_0 = \frac{1}{5}(-363.06 - 300.77 - 327.56 - 219.71 - 451.21)$$

$$= -332.46$$

$\therefore \theta_1 = 8.66 - 0.1(-69.94) = 15.65$

$\theta_2 = 8.97 - 0.1(-73.16) = 16.29$

$\theta_0 = 36.94 - 0.1(-332.46) = 70.19$

Iteration 3:

$$h_\theta(x) = 70.19 + 15.65\,x_1 + 16.29\,x_2$$

Row 1 : 70.19 ; Error = $70.19 - 400 = -329.81$

Row 2 : $70.19 + 15.65(-0.89) = 56.26$; Error = $56.26 - 330 = -273.74$

Row 3 : $70.19 + 15.65(0.52) = 78.33$; Error = $78.33 - 369 = -290.67$

Row 4 : $70.19 + 15.65(-1.21) + 16.29(-1.58) = 25.31$; Error = -206.49

Row 5 : $70.19 + 15.65(1.58) + 16.29(1.58) = 120.66$; Error = -395.34

$$\text{grad}_{\theta_1} = \frac{1}{5}\left[0 + (-273.7 \times -0.89) + (-290.7 \times 0.52) + (-206.5 \times -1.21) + (-395.3 \times 1.58)\right]$$

$$= -56.46$$

$$\text{grad}_{\theta_2} = \frac{1}{5}\left[0 + 0 + 0 + (-206.5 \times -1.58) + (-395.3 \times 1.58)\right]$$

$$= -59.66$$

$$\text{grad}_{\theta_0} = \frac{1}{5}\left(-329.81 - 273.74 - 290.67 - 206.49 - 395.34\right]$$

$$= -299.21$$

$\therefore$ $\theta_1 = 15.65 - 0.1(-56.46) = 21.30$

$\theta_2 = 16.29 - 0.1(-59.66) = 22.26$

$\theta_0 = 70.19 - 0.1(-299.21) = 100.11$

$\therefore$ ~~Price = 41~~

$$h_\theta(x) = 100.11 + 21.30(x_1') + 22.26(x_2')$$

Feature 1: Living area 2200 $feet^2$

- $\mu_1 = 2104$
- $\sigma_1 \approx 568.82$

$\therefore x_1' = \frac{2200 - 2104}{568.82} \approx 0.169$

Feature 2:

- $\mu_2 = 3$
- $\sigma_2 \approx 0.633$

$x_2' = \frac{4-3}{0.632} \approx 1.582$

$\therefore$ price $= 100.11 + (21 \times 0.169) + (22.26 \times 1.582)$

$= 138.93$

$\therefore$ Predicted price $= 138.93 \times 1000$

$= \$138930$

Ans

[Ans to the Question no 1] [Alternative way]

$$\text{Hypothesis}\,(h_\theta(x)) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$$\theta = (X^T X)^{-1} X^T y$$

$$X = \begin{bmatrix} 1 & 2104 & 3 \\ 1 & 1600 & 3 \\ 1 & 2400 & 3 \\ 1 & 1416 & 2 \\ 1 & 3000 & 4 \end{bmatrix}$$

$$y = \begin{bmatrix} 400 \\ 330 \\ 369 \\ 232 \\ 516 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2104 & 1600 & 2400 & 1416 & 3000 \\ 3 & 3 & 3 & 2 & 4 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 10520 & 15 \\ 10520 & 25931712 & 32256 \\ 15 & 32256 & 47 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1847 \\ 4387280 \\ 5673 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 2.23 & -0.00047 & -0.89 \\ -0.00047 & 0.00000032 & -0.000018 \\ -0.89 & -0.000018 & 0.47 \end{bmatrix}$$

$$\therefore \theta = (X^T X)^{-1} X^T y$$

$$= \begin{bmatrix} -41.10 \\ 0.05697 \\ 96.88 \end{bmatrix}$$

$\therefore$ price = $-41.10 + 0.05697$ (Living Area) $+ 96.88$ (Bedrooms)

Price prediction of 4 bedrooms and 2200 $feet^2$ of Living Area.

$$\therefore price = -41.10 + 0.05697\,(2200) + 96.88\,(4)$$
$$= -41.10 + 125.33 + 387.52$$
$$= 471.75$$

$\therefore$ Predicted price $= 471.75 \times 1000$

$= \$\,471750$

(Ans)

[Ans to the question no 2]

Ex-1

Low = 0
High = 1

Let's scale the area by dividing by 1000.

$\therefore x^{(1)} = 1.0$ (1000 sqft), $y^{(1)} = 0$
$x^{(2)} = 1.5$ (1500 "), $y^{(2)} = 1$
$x^{(3)} = 2$ (2000 "), $y^{(3)} = 1$

Let,

$w = 0$
$b = 0$
$\alpha = 0.1$

$\therefore \hat{y} = wx + b$

loss function gradient:

$$\frac{\partial J}{\partial w} = \frac{1}{m} \sum (\hat{y} - y) x$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum (\hat{y} - y)$$

$m = 3$

Iteration 1:

$$\hat{y} = 0 * x + 0$$

$\hat{y} = 0$, Error $(0-0) = 0$
$\hat{y} = 0$, Error $(0-1) = -1$
$\hat{y} = 0$, Error $(0-1) = -1$

$$grad_w = \frac{1}{3}[0 \times 1 + (-1) \times 1.5 + (-1) \times 2]$$
$$= \frac{-3.5}{3} \approx -1.167$$

$$grad_b = \frac{1}{3}[0 + (-1) + (-1)]$$
$$= \frac{-3}{3} \approx -0.667$$

$$w_{new} = w - \alpha (grad_w)$$
$$= 0 - 0.1(-1.167)$$
$$= 0.117$$

$$b_{new} = b - \alpha(grad_b)$$
$$= 0 - 0.1(-0.667)$$
$$= 0.067$$

Iteration 2:

$$\hat{y} = 0.117x + 0.067$$

$x=1$, $\hat{y} = 0.117(1) + 0.067 = 0.184$, Error $= 0.184 - 0 = 0.184$

$x=1.5$, $\hat{y} = 0.117(1.5) + 0.067 = 0.243$, Error $= 0.243 - 1 = -0.757$

$x=2$, $\hat{y} = 0.117(2) + 0.067 = 0.301$, Error $= 0.301 - 1 = -0.699$

$$grad_w = \frac{1}{3}\left[0.184 \times 1 + (-0.757) \times 1.5 + (-0.699) \times 2\right]$$

$$= \frac{1}{3}\left[0.184 - 1.135 - 1.398\right]$$

$$= -0.783$$

$$grad_b = \frac{1}{3}\left[0.184 + (-0.757) + (-0.699)\right]$$

$$= -0.424$$

$$w_{final_2} = 0.117 - 0.1(-0.783) = 0.195$$

$$b_{final_2} = 0.067 - 0.1(-0.424) = 0.109$$

For 1400 sq. feet,

$$y = 0.195\left(\frac{1400}{1000}\right) + 0.109$$

$= 0.382$; closer to 0 than 1

$\therefore$ So, prediction of the price level is low.

Iteration 3:

$$\hat{y} = 0.195x + 0.109$$

$x=1$, $\hat{y} = 0.195(1)+0.109 = 0.304$, Error: $0.304-0 = 0.304$

$x=1.5$, $\hat{y} = 0.195(1.5)+0.109 = 0.402$, Error $= 0.402 - 1 = -0.598$

$x=2$, $\hat{y} = 0.195(2)+0.109 = 0.499$, Error $= 0.499 - 1 = -0.501$

$$grad_w = \frac{1}{3}\left[(0.304 \times 1) + (-0.598) \times 1.5 + (-0.501 \times 2)\right]$$

$$\approx -0.531$$

$$grad_b = \frac{1}{3}\left[0.304 - 0.598 - 0.501\right]$$

$$\cancel{\approx 0.136} \approx 0.265$$

$$w_3 = 0.195 - 0.1(-0.531) = 0.248$$

$$b_3 = 0.109 - 0.1(-0.265) = 0.136$$

For 1400 sq. ft:

Input, $x = 1.4$

$$\hat{y} = 0.248(1.4) + 0.136$$

$$= 0.347 + 0.136$$

$$= 0.483 \rightarrow \text{closer to } 0 (< 0.5).$$

$\therefore$ So, Low price

(Ans)

Ex-2

Low = 1

medium = 2

High = 3

Scaled Inputs:

x_1 (Area/1000): [2.1, 1.6, 2.4, 1.4, 3.0]

x_2 (Bedrooms): [3, 3, 3, 2, 4]

model: $\hat{y} = w_1x_1 + w_2x_2 + b$

Let,
$w_1 = 0$
$w_2 = 0$
$b = 0$

Iteration 1:

$x_1 = 2.1$, $x_2 = 3$, $\hat{y} = 0$, Error $= 0 - 2 = -2$

$x_1 = 1.6$, $x_2 = 3$, $\hat{y} = 0$, Error $= 0 - 1 = -1$

$x_1 = 2.4$, $x_2 = 3$, $\hat{y} = 0$, Error $= 0 - 2 = -2$

$x_1 = 1.4$, $x_2 = 2$, $\hat{y} = 0$, Error $= 0 - 1 = -1$

$x_1 = 3.0$, $x_2 = 4$, $\hat{y} = 0$, Error $= 0 - 3 = -3$

$grad_{w_1} = \frac{1}{5}[-2(2.1) - 1(1.6) - 2(2.4) - 1(1.4) - 3(3.0)]$

(m=5) ≈ -4.2

$grad_{w_2} = \frac{1}{5}(-2(3) - 1(3) - 2(3) - 1(2) - 3(4)]$

≈ -5.8

$grad_b = \frac{1}{5}((-2) + (-1) + (-2) + (-1) + (-3)) = -1.8$

$w_{1,1} = 0 - 0.1(-4.2) = 0.42$

$w_{2,1} = 0 - 0.1(-5.8) = 0.58$

$b_1 = 0 - 0.1(-1.8) = 0.18$

Iteration 2:

$\hat{y} = 0.42x_1 + 0.58x_2 + 0.18$

$x_1 = 2.1, x_2 = 3, \hat{y} = 0.42\times2.1 + 0.58\times3 + 0.18 = 2.08$, Error $= 2.8 - 2 = 0.8$

$x_1 = 1.6, x_2 = 3, \hat{y} = 0.42\times1.6 + 0.58\times3 + 0.18 = 2.6$, Error $= 2.6 - 1 = 1.6$

$x_1 = 2.4, x_2 = 3, \hat{y} = 0.42\times2.4 + 0.58\times3 + 0.18 = 2.9$, Error $= 2.9 - 2 = 0.9$

$x_1 = 1.4, x_2 = 2, \hat{y} = 0.42\times1.4 + 0.58\times2 + 0.18 = 1.9$, Error $= 1.9 - 1 = 0.9$

$x_1 = 3.0, x_2 = 4, \hat{y} = 0.42\times3 + 0.58\times4 + 0.18 = 3.7$, Error $= 3.7 - 3 = 0.7$

$grad_{w_1} = \frac{1}{5}[0.8\times2.1 + 1.6\times1.6 + 0.9\times2.4 + 0.9\times1.4 + 0.7\times3.0]$

$= 1.952 \approx 2.0$

$grad_{w_2} = \frac{1}{5}[0.8\times3 + 1.6\times3 + 0.9\times3 + 0.9\times2 + 0.7\times4]$

$= 2.9 \approx 3.0$

$grad_b = \frac{1}{5}[0.8 + 1.6 + 0.9 + 0.9 + 0.7]$

$= 0.98 \approx 1.0$

$w_{1,2} = 0.42 - 0.1\times(2.0) = 0.22$

$w_{2,2} = 0.58 - 0.1\times(3.0) = 0.28$

$b_2 = 0.18 - 0.1(1.0) = 0.08$

Iteration 3 :

$$\hat{y} = 0.22\,x_1 + 0.28\,x_2 + 0.08$$

$x_1 = 2.1, x_2 = 3$, $\hat{y} = 0.22\times 2.1 + 0.28\times 3 + 0.08 = 1.38$, Error $= 1.38 - 2 = -0.62$

$x_1 = 1.6, x_2 = 3$, $\hat{y} = 0.22\times 1.6 + 0.28\times 3 + 0.08 = 1.27$, Error $= 1.27 - 1 = 0.27$

$x_1 = 2.4, x_2 = 3$, $\hat{y} = 0.22\times 2.4 + 0.28\times 3 + 0.08 = 1.45$, Error $= 1.45 - 2 = -0.55$

$x_1 = 1.4, x_2 = 2$, $\hat{y} = 0.22\times 1.4 + 0.28\times 2 + 0.08 = 0.95$, Error $= 0.95 - 1 = -0.05$

$x_1 = 3, x_2 = 4$, $\hat{y} = 0.22\times 3 + 0.28\times 4 + 0.08 = 1.86$, Error $= 1.86 - 3 = -1.14$

~~$x_1 =$~~

$$grad_{w_1} = \frac{1}{5}\left[(-0.62)\times 2.1 + (0.27\times 1.6) + (-0.55)\times 2.4 + (-0.05)\times 1.4 + (-1.14)\times 3\right]$$

$$\approx -1.13$$

$$grad_{w_2} = \frac{1}{5}\left[(-0.62)\times 3 + (0.27\times 3) + (-0.55)\times 3 + (-0.05\times 2) + (-1.14\times 4)\right]$$

$$\approx -1.46$$

$$grad_b = \frac{1}{5}\left[-0.62 + 0.27 - 0.55 - 0.05 - 1.14\right]$$

$$\approx -0.41$$

$$w_{1,3} = 0.22 - 0.1(-1.13) = 0.33$$

$$w_{2,3} = 0.28 - 0.1(-1.46) = 0.43$$

$$b_3 = 0.08 - 0.1(-0.41) = 0.12$$

FOR, 4 Beds, 2200 feet^2 of Living Area,

$$\hat{y} = 0.33x_1 + 0.43x_2 + 0.12$$

$$= 0.3(2.2) + 0.4(4) + 0.12$$

$= 2.566$; which is closer to 3.

So, High price

(Ans)

[Another Way]

Ex-2

Low = 1

Medium = 2

High = 3

$$X = \begin{bmatrix} 1 & 2104 & 3 \\ 1 & 1600 & 3 \\ 1 & 2400 & 3 \\ 1 & 1416 & 2 \\ 1 & 3000 & 4 \end{bmatrix} \; ; \; y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T y$$

$$X^T X = \begin{bmatrix} 5 & 10520 & 15 \\ 10520 & 24773856 & 32312 \\ 15 & 32312 & 47 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 4 \\ 9848 \\ 11 \end{bmatrix}$$

$$\therefore (X^T X)^{-1} X^T y = \begin{bmatrix} -1.2 \\ 0.0007 \\ 0.45 \end{bmatrix}$$

$$\hat{y} = 0.0007 \text{ (Area)} + 0.45 \text{ (Bedrooms)} - 1.2$$

For 2200 feet^2 of living area with 4 Bedrooms,

$$\hat{y} = 0.0007 \times 2200 + 0.45 \times 4 - 1.2$$

$$= 2.14$$

↳ High

[Ans to the question no 3]

a)

Yes, preprocessing is required. Feature scaling (normalization or standardization) is necessary because the input features have different ranges, which would otherwise negatively affect training and convergence.

Reasoning:

1) The feature values x_1, x_2, x_3 are on very different scales.

$x_1 \approx 3 - 10$

$x_2 \approx 124 - 398$

$x_3 \approx -172$ to -112

2) Large scale differences cause:

i) slow convergence in gradient descent

ii) One feature dominating others.

3) Two different targets:

i) $y \rightarrow$ regression (cotinuous)

ii) $z \rightarrow$ classification (binary) b)

hypothesis fuction, $h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$

cost function, $J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

MSE
↓
Mean Squared Error

| m = number of samples
| $y^{(i)}$ = true output
| $h_\theta(x^{(i)})$ = predicted output

c)

model (i) is better. Since it avoids excessively large coefficients and is more stable given the feature scale differences.

model (ii) assigns much larger weight to x_1 and slightly larger weight to x_3. Larger coefficients can lead to overfitting.

Predictions:

Sum of squared error
$= (1.853-0.4)^2 + (3.572-0.32)^2 + (2.624+0.19)^2$
$= 20.605$

$\therefore$ cost $= \frac{1}{2 \times 3} \times 20.605$
$= 3.434$

$\hat{y}_1 = 0 + 0.1 \times 10 + 0.008 \times 124 + 0.001 \times (-139) = 1.853$

$\hat{y}_2 = 0 + 0.1 \times 5 + 0.008 \times 398 + 0.001 \times (-112) = 3.572$

$\hat{y}_3 = 0 + 0.1 \times 3 + 0.008 \times 312 + 0.001 \times (-172) = 2.624$

$\hat{y}_1 = 0 + 0.5 \times 10 + 0.008 \times 124 + 0.002 \times (-139) = 5.714$

$\hat{y}_2 = 0 + 0.5 \times 5 + 0.008 \times 398 + 0.002 (-112) = 5.460$

$\hat{y}_3 = 0 + 0.5 \times 3 + 0.008 \times 312 + 0.002 (-172) = 3.652$

Sum of squared error = $(5.714-0.4)^2 + (5.460-0.32)^2$

$+ (3.652+0.19)^2$

$= \cancel{14.761}\ 69.42$

$\therefore$ cost, $J = \frac{1}{2\times3} \times 69.42$

$= 11.57$

cost of model (i) < cost of model (ii)

d) Forward pass for Neural Network with One hidden layer (4 neurons)

Assuming input features are preprocessed to x_1', x_2', x_3'.

The network has:

- Input layer: 3 neurons (for x_1', x_2', x_3')
- Hidden Layer: 4 " with activation function g (ex: ReLU or tanh)
- Output layer: 2 neurons (for binary classification using softmax)

Let,

$W^{(1)} \in \mathbb{R}^{4\times3}$ be the weight matrix from input to hidden layer.

$b^{(1)} \in \mathbb{R}^{4}$ be the bias vector for hidden layer.

$W^{(2)} \in \mathbb{R}^{2\times4}$ be the weight matrix from hidden to output layer.

$b^{(2)} \in \mathbb{R}^{2}$ be the bias vector for output layer.

✓ Forward pass equations:

Input vector, $x = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix}$

1. Hidden layer input: $z^{(1)} = W^{(1)} \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} + b^{(1)}$

2. Hidden layer output: $a^{(1)} = g(z^{(1)})$

3. Output layer input: $z^{(2)} = W^{(2)} a^{(1)} + b^{(2)}$

4. Output layer (softmax):

$$\hat{y}_1 = \frac{e^{z_1^{(2)}}}{e^{z_1^{(2)}} + e^{z_2^{(2)}}} \quad , \quad \hat{y}_2 = \frac{e^{z_2^{(2)}}}{e^{z_1^{(2)}} + e^{z_2^{(2)}}}$$

$\downarrow$ for class 0 $\qquad$ $\downarrow$ for class 1

[Ans to the question no 1]

a) Linear activation functions are not ideal because they prevent neural networks from learning complex non-linear relationships. Non-linear activation function like tanh and sigmoid allow neural networks to model complex decision boundaries.

b) The neural network has;

Input layer : 4 neurons (flattened 2x2 image)

Hidden layer : 2 neurons , activation = tanh

Output layer : 1 neuron , activation = sigmoid

Input to hidden:

$$W_{input-hidden} \ (2\times4) = \begin{bmatrix} 0.1 & 0.5 & 0.3 & 0.1 \\ 0.2 & 0.4 & 0.09 & 0.3 \end{bmatrix}$$

$$b_{hidden} \ (2\times1) = \begin{bmatrix} -0.32 \\ -0.28 \end{bmatrix}$$

Hidden to Output:

$$W_{hidden-output} \ (1\times2) = [0.5 \quad 0.9]$$

$$b_{output} \ (1\times1) = [0.5]$$

e)

Input Image (2×2) :

$$\begin{bmatrix} 0.7 & 0.8 \\ 0.6 & 0.9 \end{bmatrix}$$

flattened input vector (row-major):

$$x = \begin{bmatrix} 0.7 & 0.8 & 0.6 & 0.9 \end{bmatrix}^T$$

Hidden layer Input:

$$z_{hidden} = w_{input\text{-}hidden} \times x + b_{hidden}$$

$$= \begin{bmatrix} 0.1 \times 0.7 + 0.5 \times 0.8 + 0.3 \times 0.6 + 0.1 \times 0.9 \\ 0.2 \times 0.7 + 0.4 \times 0.8 + 0.09 \times 0.6 + 0.3 \times 0.9 \end{bmatrix} + \begin{bmatrix} -0.32 \\ -0.28 \end{bmatrix}$$

$$= \begin{bmatrix} 0.74 \\ 0.784 \end{bmatrix} + \begin{bmatrix} -0.32 \\ -0.28 \end{bmatrix}$$

$$= \begin{bmatrix} 0.42 \\ 0.504 \end{bmatrix}$$

Hidden layer output:

$$a_{hidden} = \tanh(z_{hidden})$$

$$= \begin{bmatrix} \tanh(0.42) \\ \tanh(0.504) \end{bmatrix}$$

$$\approx \begin{bmatrix} 0.397 \\ 0.465 \end{bmatrix}$$

Output layer input:

$$z_{output} = W_{hidden-output} \times a_{hidden} + b_{output}$$

$$= \begin{bmatrix} 0.5 & 0.9 \end{bmatrix} \times \begin{bmatrix} 0.397 \\ 0.465 \end{bmatrix} + \begin{bmatrix} 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 \times 0.397 + 0.9 \times 0.465 \end{bmatrix} + \begin{bmatrix} 0.5 \end{bmatrix}$$

$$= 1.117$$

Output layer activation (sigmoid):

$$\hat{y} = \sigma(1.117) = \frac{1}{1 + e^{-1.117}} \approx 0.753$$

Since, $\hat{y} > 0.5$; the image is classified as light (label 1)

[Lecture-11]

Ex-1

Layer	Input Dimensions	Filter size	# Filters or, # Neurons	Padding	Output Dimensions	# Params
conv 1	128*128*3	5*5	32	2	128*128*32	2432
Max Pool 1	128*128*32	2*2	—	0	64*64*32	0
conv2	64×64×32	3*3	64	0	62*62*64	18496
maxpool2	62*62*64	2*2	—	0	31*31*64	0
Flatten	31*31*64	—	—	—	61504	0
FC	61504	—	128	—	128	7872640
FC(output)	128	—	4	—	4	516

$$\text{Output dimension} = \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1$$

~~maxpo~~

pooling ⇨ $\frac{W + 2P}{F}$

convolutional layer, #params $= \{(K_w \times K_h \times C_{in}) + 1\} \times C_{out}$

FC , " $= (\#\text{input units} \times \#\text{output units}) + \#\text{output units}$

So, the output layer has 4 neurons, indicating that the network is designed for a classification task with 4 classes.

Part2:

The receptive field (RF) is calculated using the formula:

$$RF_l = RF_{l-1} + (K_l - 1) \times \prod_{i=1}^{l-1} s_i$$

1. Receptive field of a neuron in conv2

Input → conv1 → maxpool 1 → conv 2

i) conv1 (5×5, stride 1):

- RF = 1 + (5 - 1) × 1 = 5
- current total stride = 1

ii) max pool 1 (2×2, stride 2):

- RF = 5 + (2 - 1) × 1 = 6
- current total stride = 1×2 = 2

iii) conv2 (3×3, stride 1):

- RF = 6 + (3 - 1) × 2 = 6 + 4 = 10

∴ The receptive field dimension is (10 × 10)

2. Receptive Field if maxpool1 was removed

Input $\rightarrow$ conv 1 $\rightarrow$ conv 2

i) conv1 (5×5, stride 1):

- RF $= 1 + (5-1) \times 1 = 5$
- current total stride $= 1$

ii) conv2 (3×3, stride 1):

- RF $= 5 + (3-1) \times 1 = 5 + 2 = 7$

So, if maxpool 1 was removed, the dimensions would be (7×7).